

Techno-economic analysis of off-grid PV-Diesel power generation system for rural electrification: A case study of Chilubi district in Zambia

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ABSTRACT

The study explores the techno-economic feasibility and viability of a Photovoltaic-Diesel Hybrid system for rural electrification in sub-Saharan Africa with a case study of Chilubi island, a remote district without access to electricity in the Northern Province of Zambia. Using HOMER (Hybrid Optimization of Multiple Electric Renewables) Pro software, the best and most feasible technical solutions through different hybrid system configurations, combinations and the district's rate of access to electricity were considered based on the least Levelized Cost of Energy (LCoE) and life cycle costs of the project. The results show that operating diesel generators as stand-alone is not economically sustainable and has a high LCoE. Influencing factors include variability in diesel pump prices, high fuel transportation costs, high cost of operation and maintenance, among other factors of concern. 100% photovoltaic (PV) with a battery system gave the lowest LCoE. However, the initial capital cost of solar energy projects in Zambia is relatively high compared to the equivalent diesel-based plants, as shown herein. It explains why diesel power plants are favoured for off-grid settlements. On the hand, the low operational cost and LCoE of PV power plants favour rural districts as they offset the high initial capital costs. Additionally, the continued downward trend in the cost of PV installations per kWp has opened discussions among policymakers and energy planners in Zambia to favour rural electrification with renewable energy-based power generation. This study contributes to this discussion.

1. Introduction

The global energy demand is high and is projected to increase further [1]. The high demand and projected increase are due to the need to satisfy the universal basic energy needs of the world population, sustain the lifestyles of developed countries and develop and industrialise low-income countries [2]. This projected trend of energy use and the dependency of the world's energy economy on fossil fuels has raised concerns due to increased greenhouse gas emissions, which are the primary cause of global warming [3]. Much of the increase in energy demand will occur among countries outside the OECD (Organization for Economic Cooperation and Development) and can be attributed to economic development activities. Recent forecasts suggest that by 2040, the energy demand for non-OECD countries will be 71% higher than in 2012 [4]. In sub-Saharan Africa, taking the Eastern Africa Power Pool

(EAPP) and the Southern African Power Pool (SAPP) as examples, electricity generation will need to be tripled by 2030 to satisfy the general demand [5]. In most cases, it will not guarantee access for all the population, especially in rural areas.

In [6], the International Energy Agency (IEA) is referred to and identifies off-grid small-scale electricity generation as one of the most appropriate solutions for rural electrification and suggests that these may serve as a building block for future power grids with distributed generation. Besides, the forecast [7,8] shows that 60% of needed electricity for universal access to energy will come from off-grid systems, and about 90% of that will be based on Hybrid Renewable Energy Systems (HRESs). Literature (see Ref. [6] and the references therein) offers several reasons for the primacy of rural electrification with mini-grid small-scale generation for developing countries. The current study adopts from Ref. [6], the taxonomy for off-grid systems with specifics on

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hybrid micro-grid systems as having a distribution network, multiple consumers and energy sources. Much of the research conducted consider PV hybrid micro-grid systems under the coverage of Asian countries [6, 9]. In other cases [10], the addition of wind energy which is also a mature technology is sought in many developed and some developing countries. The capital cost on wind energy is normally higher and the potential of wind energy for a region of interest needs to be quantified and models verified. In many regions in sub-Saharan Africa, for example in Zambia, the resource modelling of the wind energy resource is ongoing, and many countries are yet to establish a utility-scale wind farm. In Zambia, plans are underway on potential sites for wind farms with measurement equipment and instruments installed for accurate modelling.

In Zambia, many rural and some areas in urban setups have little to no access to electricity. Estimates show that about 67% of people living in urban Zambia have access to grid electricity, and about 4% in rural areas [11]. The Indaba Agricultural Policy Institute reports that about 81% of households in rural areas use firewood as the primary energy source for cooking, 16% use charcoal, and only 3% use electricity [12]. To increase rural electrification, the Zambian government established the Rural Electrification Authority (REA), and 1217 rural growth centres were identified in 2006 and earmarked for electrification by 2030 using different energy technologies, according to Ref. [13]. Since commencement, the Ministry of Energy reported that about 7.4% of rural households had been electrified with off-grid systems. Like many other African countries, the program has faced many challenges, including financing, as it is aid-dependent [14].

In this study, we explore the feasibility and potential of PV-diesel hybrid systems for rural electrification in Zambia. The study investigates integration of PV (photovoltaic) with diesel generators for a micro-grid power system to increase local access to electricity, power reliability and system performance in Chilubi, a rural district in the Northern part of Zambia (Northern Province). To the authors' knowledge, there are a few feasibility studies [6,15] on hybrid microgrids for rural electrification in sub-Saharan Africa, particularly in southern Africa. Authors in Ref. [5] emphasise the importance of strategic siting and regional planning for the successful implementation of energy projects. Therefore, models and frameworks of rural electrification developed in other regions may not be as effective due to social, political, and economic differences. For example, economic aspects of grid expansion and a lack of economic activities in rural areas hinder rural electrification projects in Zambia and in cases where there is a greater need for electricity, diesel generation plants are used. Even with diesel generation plants, the fuel cost will differ according to the location of the generation site because of the transportation and storage costs of fuel. The social and political aspects, including high urbanisation, have shifted and offset rural electrification plans due to increased population and demand for grid electricity in urban areas [16]. In addition, the decision makers' goals and plans to offset the energy deficit in the region seem hydropower centric [17–19], and in some cases, there is a drive toward fossil fuels, which is being considered an under-tapped resource [20].

Despite the history of drought vulnerability in the regions, reduced water levels in major hydro reservoirs have become common and consequently affect power generation [20]; hydropower generation systems still dominate electricity generation and energy planning in Zambia and across east and southern Africa. This centralised energy paradigm has proven costly to operate, develop and extend the power grid to areas far from the generation facilities [5]. Consequently, there is limited access to the national power grid, and for those with access, the cost of electricity is more than twice as high as in other developing countries; thus, governments subsidise the cost of electricity [21]. Electric supply expansion/access is still a challenge needing urgent attention, particularly in remote/rural areas, which are the most affected [22]. It is not unique to Zambia or Southern Africa; literature recognises that multiple dimensions hamper universal access to energy in developing countries: lack of reliable supply of energy for a large

population due to insufficient installed capacity, poor energy infrastructure management and difficulties in attracting investment in the energy sector among others [5,6,23,24].

To meet sustainable development goals and ensure universal access to affordable, reliable, and sustainable modern energy for all, efforts need to be doubled on rural electrification in most regions in sub-Saharan Africa, especially in regions like Zambia where access to electricity in rural areas is less than 5% [12]. As earlier stated, off-grid small-scale electricity generation are the most appropriate solutions for rural electrification [6]. However, system instability increases with increase in the share of renewable sources due to the intermittent nature of energy sources, thus integration of storage is encouraged to increase stability and inertia of the power system. Including storage increases system complexity and the capital cost of microgrids [6,9]. Alternative forms of microgrid incorporate solar-pumped hydro system [25]. In this system, the investment costs are even higher as the pumped hydro system needs additional components to that of the traditional PV system. The components needed are pump(s), at least two reservoirs, penstock, turbine/generator arrangement and piping materials [26,27]. Energy storage using pumped hydro is mature and an established technology with utility-scale application [26–28]. However, it is characterized by high capital costs with low profitability, which is a barrier to its development in certain cases for utility-scale [29]. In addition, it is very dependent on local topography and landscapes. For sites with hydro pumping storage potential, an hybrid system with pumped hydro becomes far more cost-effective in the long run than using a PV-battery combination [30]. The current study does not consider pumped hydro since the regional topography, despite being surrounded by water, is flat and does not favour pumped hydro power storage.

The current study contributes to the general framework of rural electrification with hybrid micro-grids that can be implemented to supply electricity to rural communities in Zambia. More importantly, it contributes to the definition of a reference for implementing hybrid microgrids for rural electrification. Additionally, this study will offer insight to government policy and decision-makers on options, among other things, for rural electrification when it comes to implementing renewable energy strategies.

The paper is arranged into six sections. It begins with Section 1, the introduction and outline of microgrids for rural electrification. In Section 2, the study area in this paper is described. In Section 3, a presentation of the method and modelling is described. The simulation results are presented in Section 4. In Section 5, discussions of the obtained results are presented. The conclusions are described in Section 6.

2. Study area-Chilubi district

Chilubi district, situated about 900 m above sea level, is in Northern Province of Zambia. Topographically, the district is partly on mainland Zambia, and an island on Lake Bangweulu which borders Kasama district on the eastern side, Samfya district on the western side and Luwingu district on the northern side. It is one of the most remote towns in Zambia, and the road network is underdeveloped. Access to the island and mainland part of the district is mainly by water vessels from Samfya district via Lake Bangweulu and road via Luwingu district, respectively [31]. The district is approximately 919 km by road from Lusaka (Zambia's capital city) via Luwingu town. According to the national statistical population census summary report of 2010 [32], Chilubi had a population of about 81,248 and was projected to grow to about 96,703 by 2019 [33]. The Living Conditions Monitoring Survey (LCMS) conducted in 2015 [34] show that the average household family size in rural areas in Zambia is about 5.2, which is close to the value obtained (4.9) when the population of Chilubi is extrapolated against the reported number of households (about 16,716) in 2010 [35]. Today, the households are estimated to be about 19 800 based on population projections for 2019, and none are connected to the national electric grid.

There are no significant economic activities, if any minimal albeit

minor small-scale farming, as residents mainly depend on subsistence farming (e.g., cassava farming) and mainly on fishing activities [36,37]. The whole mainland part of the district has no access to electricity, and residents resolve to meet their basic energy needs by burning biomass for cooking and using kerosene for lighting at night. Those engaged in fisheries use small diesel generators for refrigeration needs before transporting the goods. Chilubi district has a mapping of resources in terms of energy potential from solar irradiation and wind measurements [38]. Details of the area and the solar power potential are described in Section 2.1.

2.1. Energy resources mapping

A strategic siting and successful implementation of energy projects emphasise utilising local energy resources as it minimises costs on energy infrastructure, transmission of power associated with long distances, maintenance and operation [5]. The case study herein implies utilising local renewable resources to help sustain the environment. A national campaign was recently conducted to map energy resources across the country; solar has the highest potential in most parts of the Western, and some parts of Northern Provinces (for the case study, it was the only viable option), and winds speeds are below generation limits and areas with good enough wind speeds are located more than 300 km from the district. Hydropower potential is very low as the landscape is flat even though the district is close to a lake [39,40]. Additionally, solar is considered the most important source of energy supply which is inexhaustible and permanent [41]. With 2000–3000 average hours of sunshine per year, Zambia has a high potential for solar energy [42].

Accurate meteorological data and solar resources are necessary ingredients for solar plant operations. However, Chilubi has no meteorological weather station. Therefore, solar irradiation potential and assessment for Chilubi were done using SolarGis [38], which accounts for data at a station in Misamfu in Northern Province. The average daily global horizontal irradiation was 5.54 kWh/m², and the yearly average temperature ranged between 18.37 °C and 28.2 °C [38].

ZESCO Limited, a state-owned and largest power company, producing about 80% of the electricity consumed in Zambia, install and operate diesel power plants for off-grid power solutions in rural areas on a need basis. With solar energy development, the drive has been toward integrating or replacing diesel power plants with solar energy. Therefore, this study considers diesel fuel and solar PV energy resources.

2.2. Needs assessment

As of 2018 [31], only the island part of the district had been connected to the national grid, and supply is only to the government district administrative centre. However, every household in the district remains unconnected. To foster development in the area and boost the quality of life and economic activities, particularly agriculture and fisheries, the residents need to have access to electricity. Connecting clinics, schools, and markets can increase health care, enhance education, reduce illiteracy, and increase trading hours.

Most houses in the district have grass-thatched roofs and depend on modified kerosene lamps for artificial lighting, which pose a danger to human health and life. Since there are no big commercial industries (even though the district has considerable potential for fisheries), the electric load would be relatively low; therefore, connecting all the households in the district may not stress the grid, but there is a lack of economic incentive. Like most remote areas [43], Chilubi will require a low supply of electricity for lighting and electrical appliances for the residential units.

This study developed an average load profile per connection based on data obtained. Reference was also made to Refs. [44,45]. The illustration of the average load profile is shown in Fig. 1.

The average daily load profile in Fig. 1 shows the peak value at 7 p.m. The load is fairly distributed during the daytime. Due to the absence of

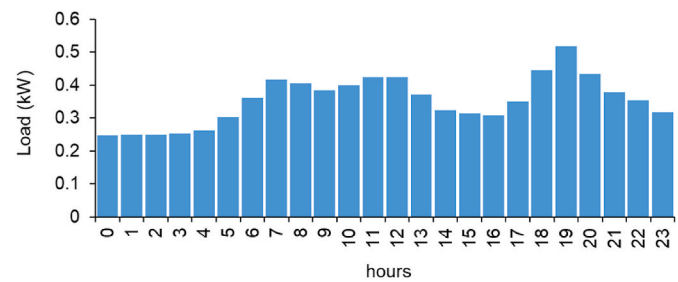


Fig. 1. Average daily load profile per connection.

the national grid, most of such areas have been supplied using a Diesel-powered micro-grid network. Another example of such an area is given in Ref. [45].

3. Method and modelling

The paper is based on modelling and mathematical application to evaluate the feasibility of renewable energy implementation for rural areas in Zambia. The cost elements of the four evaluated scenarios are based on the assessment to supply power to residents in the Chilubi district while accounting for future government targets for rural electrification.

HOMER software, a tool for designing and planning HRESs, was used to model and simulate four different scenarios (discussed later) to determine the best and most feasible technical solution. It models behaviour, life cycle costs and physical configuration of hybrid micro-power systems. The software allows the designer to evaluate the design options for grid-connected and off-grid power systems for distributed generation, stand-alone, and remote applications [46]. Additionally, the software allows the designer to compare many different design options while considering their technical and economic merits. To determine the technical feasibility and entire applicable life cycle costs, the software models the performance of a hybrid system configuration on an hourly basis for the entire year. Additionally, it requires actual local input data in the region for energy resources and costs for setting up and operating a power plant for more realistic and applicable results for that specific region. The modelling study cases are discussed in the following sections.

3.1. Modelling of study cases

Three different load profiles were evaluated to analyse and select the better optimized and technically feasible solution. The current load profile would not account for future growth. The solution for any of the three different loads will account for the area's current load and future growth. The load profiles were defined by the level of percentage of household connections to the local microgrid against the total number of households in the district. The load profiles are shown in Fig. 2.

The profiles in Fig. 2 are defined by: (a) A 9% connection rate in the district was evaluated. The current rural electrification rate is at 7%. According to the Rural Electrification of Zambia (REA), the rate is expected to rise to 9% by 2030. Therefore, 9% was selected as the first scenario to consider and shown in Fig. 2. (b) A 51% connection rate based on the Zambian government's target for rural electrification by 2035, and (c) A 90% connection rate. The electrification of rural areas such as Chilubi is expected to encourage the growth of the economy in the area. When such happens, the growth rate in rural areas will get closer to that of Urban areas. The government's target for urban areas is a growth of 90%. Therefore, the rate may grow faster for an area such as Chilubi Island. In this condition, the system was compared against the government's target for urban electrification by 2036 [47,48].

The establishment of electricity supply will cause future load profiles for Chilubi Island, as shown in Fig. 2. The projections are the products of

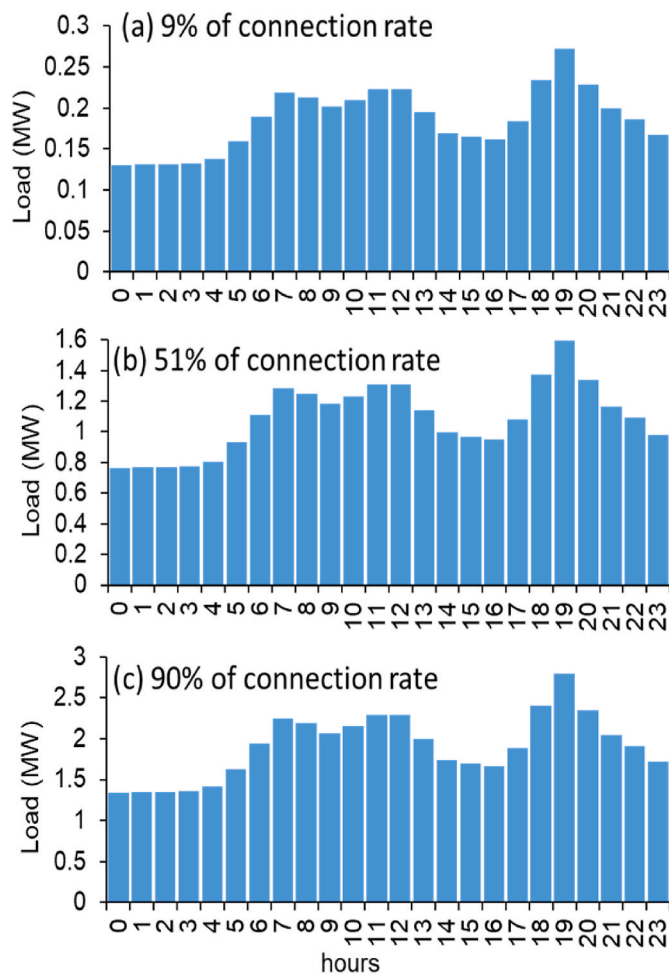


Fig. 2. Average daily load profile for different connection rates.

connection rates in the district. The applied load profile can be extended for other rural setups with similar lifestyles and economic activities. The average daily load profiles are developed based on the district's connection rate and future expected growth concerning the estimated 19800 households [31,34,40].

An optimiser in the applied software was used to determine the most feasible system capacity for the given load profile while accounting for all sensitivity cases and adjusting each component according to the load profile to obtain the LCoE. In this study and based on government conditions to favour renewable energy sources, a discount rate of 6% was used. Since, for system stability, the software increases the peak power of the load profiles by 30% (the load scaled annual average is increased by 30% of the load average baseline), no random variability was considered to avoid duplicity in increasing the peak load.

Table 1 summarises the system components and specifications used for modelling in HOMER. Cost specifications for each system component are discussed in Sections 3.1.1 to 3.1.4. The modelling costs are average values from quotations and costs obtained from local suppliers in Zambia. They account for all local taxes and transport costs to the site.

3.1.1. Scenario A – Diesel generator supply

This scenario modelled a diesel generator (DG) as the only electricity-generating source. It is a typical power plant operated in a few rural areas in Zambia by the utility company (ZESCO) to cover the load. Herein, a diesel generator is modelled to always cover the energy needs of the microgrid for both daytime and evening loads. The general layout is shown in the schematic diagram in Fig. 3.

An auto-size genset was chosen; this automatically sizes itself to meet

Table 1

System components and specifications.

DG	Type	Autosize Genset
	Rating	Software optimized
	Lifetime	35 000
	Minimum load ratio (%)	25
	Fuel curve intercept L/hr.	6.43
	Fuel curve slope (L/hr./kW)	0.236
	Lower heating value (MJ/kg)	43.2
	Density (kg/m ³)	820
	Carbon content (%)	88
	Sulphur content (%)	0.4
Fuel (Diesel)	CO (g/L)	16.5
	Unburned HC (g/L)	0.72
	Particulates (g/L)	0.1
	Fuel Sulphur to PM (%)	2.2
	NOx (g/L)	15.5
DG Emissions	CO (g/L)	16.5
	Unburned HC (g/L)	0.72
	Particulates (g/L)	0.1
	Fuel Sulphur to PM (%)	2.2
	NOx (g/L)	15.5
PV	Type	Generic flat plate PV
	Rating (kW)	Software optimized
	Derating factor (%)	80
	Ground reflectance (%)	20
	Tracking	No
	Panel slope	11
	Panel azimuth	180
	Temperature coefficient (%/C)	−0.5
	Operating temperature	47
	Efficiency	13
Converter	lifetime (yr.)	15
	Type	Generic system converter
	Rating (kW)	Software optimized
	Relative capacity (%)	100
	Efficiency (%)	95
	lifetime (yr.)	15
Batteries	Type	Lithium-Ion Battery
	Nominal voltage (V)	600
	Nominal capacity (kWh)	100
	Nominal capacity (Ah)	167
	Round trip efficiency (%)	90
	Maximum charge current (A)	167
	Maximum Discharge current (A)	500
	Initial state of charge (%)	100
	Minimum state of charge (%)	20
	Throughput (kWh)	300 000
	lifetime (yr.)	15

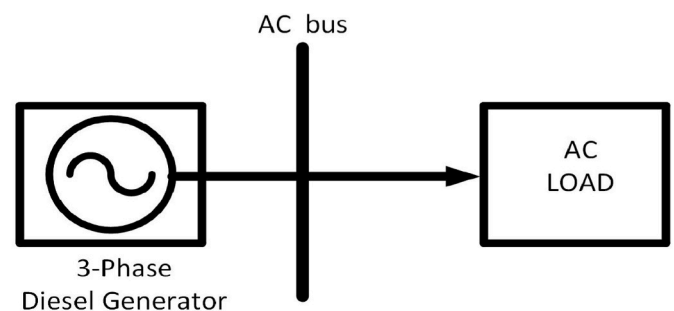


Fig. 3. Diesel stand-alone system.

the load and adjusts its fuel curve to match the size. The cost for the generator was for each kW of the capacity: The capital cost was \$250, replacement cost was set at 40% of the installation cost, and operation and maintenance were set at \$0.02 per operational hour. Site-specific inputs included: Diesel price per litre was \$1.25 (pump price plus transport to the site), generator lifetime of 35000 h and a minimum load ratio of 68% (ratio of the average load and the peak load).

3.1.2. Scenario B – diesel generator plus PV without storage

The primary supply of Scenario A is integrated with PV producing and covering the load during sun hours. This scenario considers that the

DG is only operated during nighttime to cover the basic demand at night and during the day when solar fails to satisfy the demand, e.g., prolonged cloud cover during the rainy season. The PV system has no storage included, and the HOMER optimiser determines the system capacity. The schematic diagram for this case is shown in Fig. 4.

A generic flat-plate PV with a lifetime of 25 years was chosen, with the capacity cost for each kW installed of \$700 as capital, \$700 for replacement and \$10 per annum for operation and maintenance. Equally, the converter capacity was estimated using the HOMER optimiser, and the cost was \$100 for capital and \$100 for a replacement for each kW of the installed capacity. No operation and maintenance costs were included. The chosen inverter had an efficiency of 95%, a lifetime of 15 years and a relative capacity of 100% and 95% for the rectifier, as shown in Table 1.

3.1.3. Scenario C – diesel generator plus PV with storage

Storage is integrated into the case of scenario B. The DG can cover the load that is not catered for by the PV and storage and during the days when there is no PV production and storage supply. A generic 100 kWh Lithium-ion battery is chosen, and a HOMER optimiser is used to determine the size (storage strings) depending on the load demand. The capital for setting up a unit was \$ 400/kWh; replacement cost was \$ 320/kWh, and an annual maintenance cost of \$ 10/kWh. The battery lifetime was 15 years with a throughput of 300 000 kWh; see Table 1 for more details. Fig. 5 shows the schematic diagram of scenario C.

3.1.4. Scenario D - 100% PV to cover the total load demand with storage

In this scenario, the full-fledged supply is covered by PV with storage. The storage covers the night-time load, while the PV provides for the daytime load. The system is set with the same cost inputs for PV and storage as in scenario C, and the HOMER optimiser is used to determine the capacity, except that the DG is not included in the system. Fig. 6 shows the schematic diagram for the scenario.

3.1.5. Choice of the four scenarios

The four scenarios were selected based on the base supply of the area when the national grid is not there. The initial investment for the utility company (ZESCO) would be a diesel generator. Other stakeholders for

national grid extension or implementation of renewable energy for rural areas are involved, the Ministry of Energy and Rural Electrification Authority (REA). Their mandate is to ensure that rural areas are connected to the national grid through extension (expensive) or solar power, increasing [49,50].

The four system scenarios were considered against each profile shown in Fig. 2 for the practical potential systems in the area, according to the Ministry of Energy and Rural Electrification Authority (REA).

3.2. Economic evaluations

The present study considered the LCoE and the Net Present Value (NPV). LCoE is the breakeven value of the price of electricity consumed [51]. LCoE was calculated using Equation (1).

$$LCoE = \frac{C_o \left[\frac{d(1+d)^n}{(1+d)^n - 1} \right] + ACOM + AC_{Fuel}}{NAED} \quad (1)$$

Where C_o is capital cost (Initial investment costs incurred at the beginning of the project, d discount rate (6% was considered), $ACOM$ is the annual cost of operations and maintenance, AC is annual fuel cost, $NAED$ is the net annual electricity delivered, and n is the project useful life (25 years was considered).

The Net Present Value (NPV), a measure of how profitable a project is, was calculated using Equation (2).

$$NPV = (B - C) \left[\frac{(1+d)^n - 1}{d(1+d)^n} \right] + \frac{S}{(1+d)^n} - C_o \quad (2)$$

Where S is salvage value, if NPV is equal to zero, a discount rate known as Internal Rate of Return (IRR) comes into effect. Therefore, to invest in a rewarding project, the prevailing discount rates on the market should be higher than the IRR.

3.3. Assumptions and limitations

As of 2010, the Chilubi district had 16,716 households. However, this study is limited to load connections of 9%, 51% and 90% of the

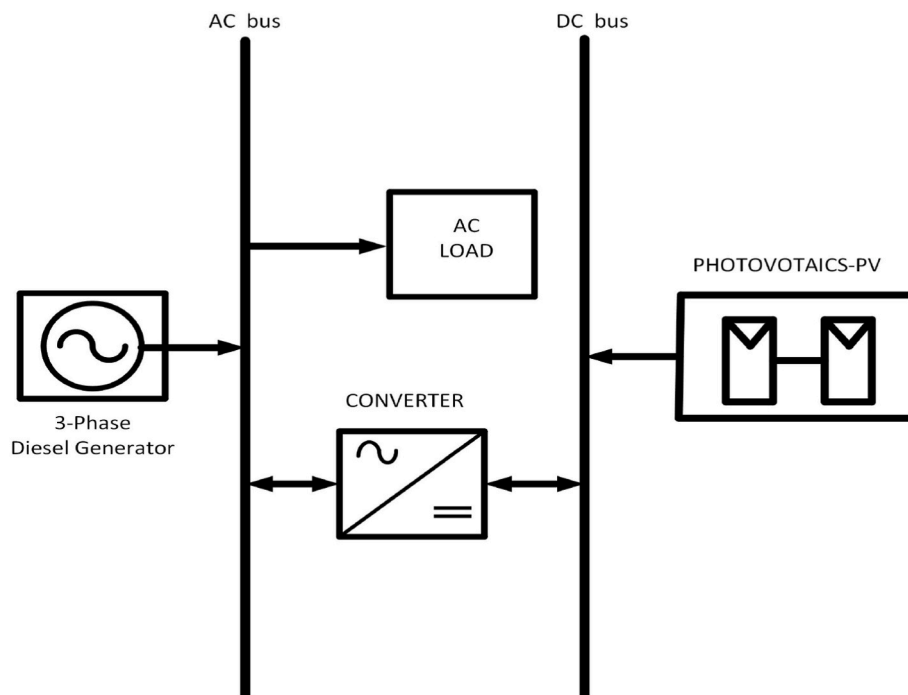


Fig. 4. The PV-Diesel hybrid system without storage.

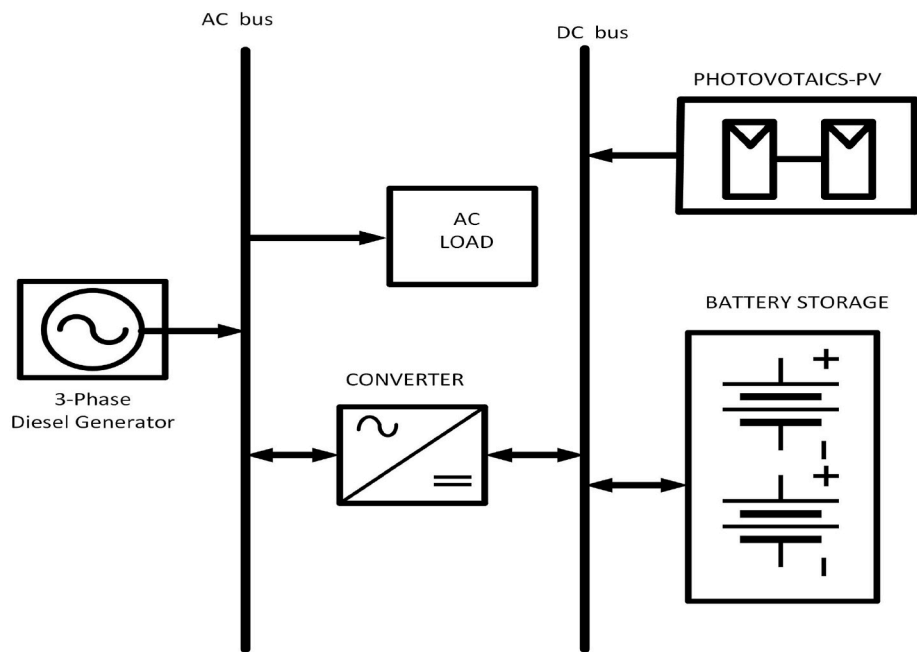


Fig. 5. PV-Diesel hybrid system with battery storage.

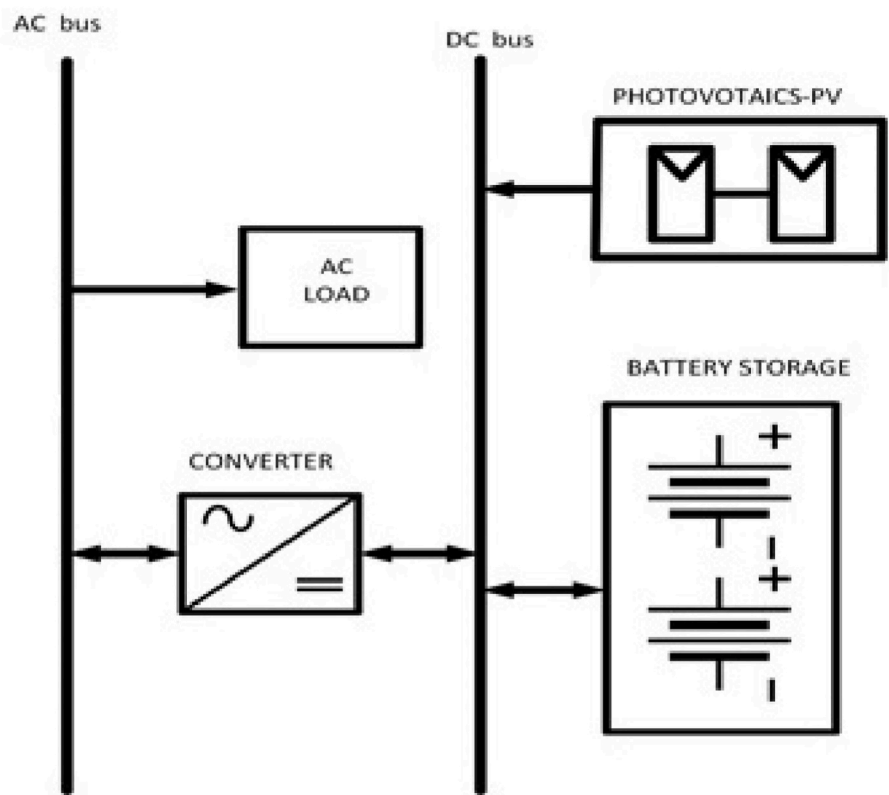


Fig. 6. 100% PV and storage to cover the total load demand.

households with current and future electrification goals. The costs used in this study are based on the quotation invoices obtained from the local companies and may change depending on the currency exchange rate. However, the results and findings of this study reflect the local trends in solar products and renewable energy products in Zambia and can be upscaled or utilised for other rural setups in Zambia and anywhere around southern Africa. Additionally, in this study, we did not account

for the cost of setting up the distribution network; thus, the analysis is limited to power generation and production equipment.

4. Results

The results obtained for each of the four system scenarios are presented in this section and discussed in the proceeding section. The

scenarios were examined and evaluated to determine the technical and economical alternative focusing on the lowest LCoE.

Table 2 shows the overall system capacities, operational costs and NPV. The basic supply scenario for 9% needs a 390 kW DG, 51% needs 2300 kW, and 90% needs a 4000 kW. In all cases the DG had 0% excess energy, DG + PV had 10%, DG + PV + Storage had 21% and PV + Storage had 44%. Integrating PV into the system produces more energy beyond the load demand, which must be curtailed for grid stability. PV capacity varies depending on the scenario, and since the system favours the lowest LCoE, excess electricity produced must be curtailed or directed to dump loads, which in this study was not considered. However, there is room for future connections or demand variability with PV, whereas operating only DG would require additional generation capacity.

Figs. 7–9 show the LCoE and investment costs against load connection scenarios at 9%, 51% and 90%, respectively. The scenario with DG as the only electricity-generating source had the lowest investment costs but the highest LCoE. The diesel generating unit's low investment costs (Initial capital) are an attractive incentive and may influence political energy investment in rural areas. For example, the 390 kW DG for a 9% connection rate has the initial capital required for the primary power supply system less than one-tenth of a million dollars, while the cost for a solution with 100% renewable penetration is almost \$4 million.

As can be observed from Figs. 7–9, the LCoE for all connection rates is almost the same for similar scenarios. At a 9% connection rate, the LCoE was \$0.373/kWh, which drops to \$0.182/kWh when the system is sustained only by PV and battery storage. The biggest challenge for DG is the fuel use; whereas the fuel consumption was 564,486 L for the 9% connection rate, an increase in connection rate leads to an increase in annual fuel consumption. The fuel consumption increased by six times and ten times for the 51% and 90% load connections, respectively. However, it has minimal effect on the LCoE when PV and storage are introduced into the system. Thus, PV and storage can be upscaled with minimal effects on the consumer's energy cost, although the investment increases.

With increased solar penetration in the system, the operation costs go down, consequently reducing the energy cost. Additionally, the consumption of more fuel entails more production of carbon dioxide or greenhouse gases into the atmosphere.

Generally, the initial cost of investment increases when solar PV is added. Increasing the connection rate to 51% and 90% leads to increased solar PV panels and hybrid inverters. Additionally, for 25 years, the generators will need to be replaced about 5 times. Introducing a high percentage of PV reduces the operation time for the DG and thus will reduce the number of times they will need replacement. The decrease in operational hours enables savings on fuel purchases for the

system. The DG operates only when there is cloud cover and at night when no storage systems are added. The addition of storage lowers the number of hours that DG operates and, consequently, the LCoE. In Scenario D, the DG units are replaced with 100% solar PV and storage to cover the entire load. The resulting scheme can also be called unconstrained supply since no fuel constraints are imposed on the system.

The initial investment needed at 9% is 3.76 million dollars. The investment needed is seven times and twelve times the lowest connection rate when the load increases. The scheme is financially high. However, the scheme would be more and very environmentally friendly than Scenario C. The LCOE is \$ 0.182/kWh at a 9% load connection and \$ 0.181/kWh for the other two connection rates. It is 51% and 47% lower than Scenario A and B. The arrangement ensures more than 100% solar power usage for the load demand through direct production stored in the battery.

5. Discussions

The study focused on the viability of the PV-Diesel Hybrid system for rural electrification in Sub-Saharan Africa, with the case study of the Chilubi district in Zambia. The scenarios of the possible microgrid to supply Chilubi Island are discussed in this section. From the simulation results, microgrids are necessary to support activities in rural areas. The cost of extending the grid (expanding high tension and distribution lines) and the low economic activities to facilitate the competitive cost of energy are prohibitive. Thus, microgrids become the best alternative in sparsely populated rural areas with low or no industrial activities. In the sub-Sahara, it is estimated that the average cost of grid extension is approximately \$ 20000/km for an 11 kV line without any other accessories [15] and approximately \$ 60000/km and \$ 300000/km for 66 kV and 330 kV lines. The huge costs associated with grid extension make the sub-Sahara's solar solutions more attractive for rural areas and a quicker way to increase electrification rates. It is, therefore, very beneficial for developing countries to consider developing microgrids compared to extending the national grid. In addition, the extension of the grid is not very attractive for the utility company since the cost of energy on the grid is highly subsidized (\$ 0.036/kWh).

In this study, to implement and achieve the government's rural goals, there is a need for capital investments for the three categories of load connections. This study helps determine the viability of the option for power production in rural areas with a similar economic level. Scenario D has the highest initial investment capital needed. The investment needed for scenario C is the second highest among the scenarios. However, in the long run, one of the two scenarios will contribute to the sustainable development agenda of low to zero carbon emissions from the power supply system (Table 1 gives system characteristics emission

Table 2
Summary of results for various connection rates and scenarios.

connection rate	Scenario	DG [kW]	PV Array [kW]	Converter [kW]	Battery [kWh]	Renewable penetration [%]	Annual Diesel [Litres]	Fuel costs [\$(million)/year]	Operating [\$(million)/year]	NPV [\$(million)]
9%	DG	390	–	–	–	0	564486	0.71	0.782	10.1
	DG + PV	390	542	199	–	25.3	437841	0.55	0.631	8.62
	DG + PV	390	1675	396	4600	92	47717	0.60	0.141	4.94
	+ BT									
	PV + BT		2346	394	5200	100	0	0	0.077	4.74
51%	DG	2300	–	–	–	0	3224469	4.03	4.48	57.9
	DG + PV	2300	1925	777	–	20.4	2089876	2.61	3.09	41.7
	DG + PV	2300	9827	2293	27100	92.1	266645	0.33	0.803	28.8
	+ BT									
	PV + BT		14370	2610	28700	100	0	0	0.47	27.8
90%	DG	4000	–	–	–	0	4464053	5.58	6.37	82.4
	DG + PV	4000	3392	1362	–	20.5	3654002	4.57	5.4	72.9
	DG + PV	4000	14370	2610	46900	91.9	479891	0.6	1.43	28.8
	+ BT									
	PV + BT		17190	4043	48900	100	0	0	0.84	27.8

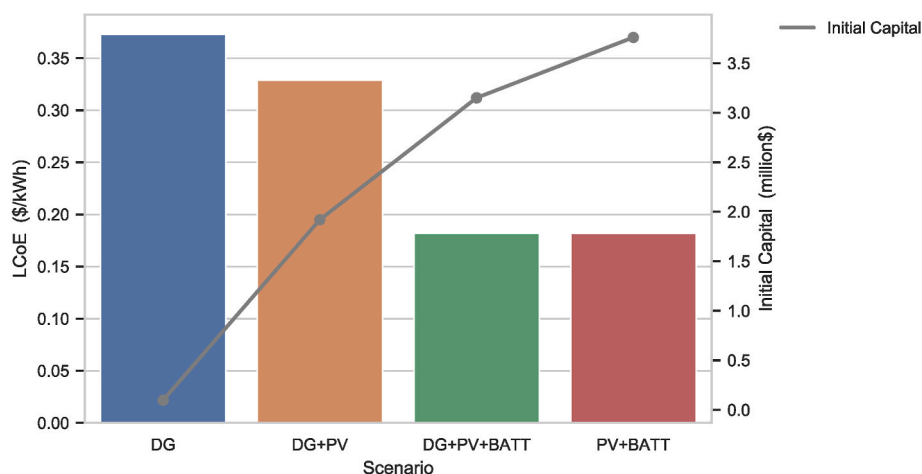


Fig. 7. LCoE and initial capital at 9% load connection.

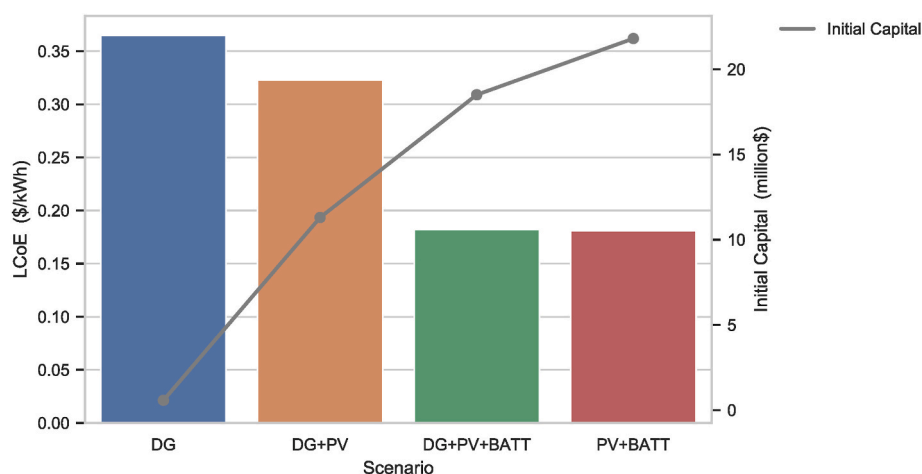


Fig. 8. LCoE and initial capital at 51% load connection.

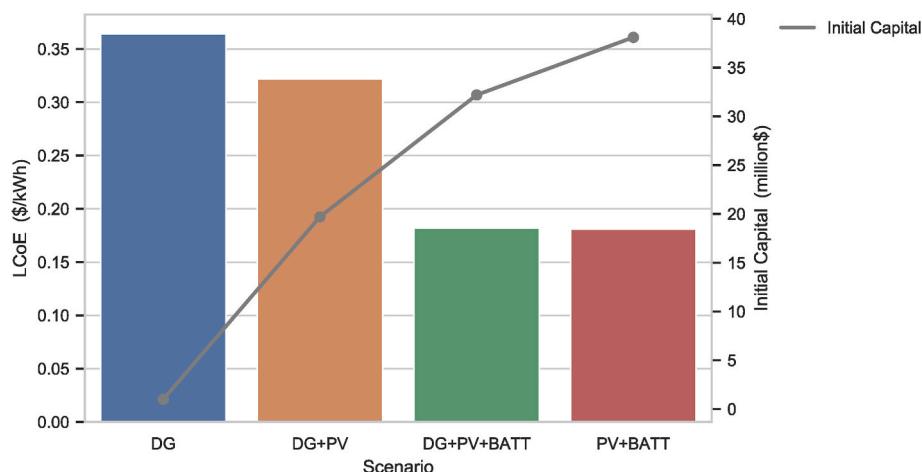


Fig. 9. LCoE and initial capital at 90% load connection.

per litre for operating the diesel generator). The two scenarios (one without use of DG and the other with reduced hours of operation the DG) agree are in line with COP26 aims of reducing/eliminating carbon emissions in power generation. The cheaper options (Scenario A and B) contribute with high carbon emissions even though they require low capital investment, they also increase operational costs.

The capital costs of PV power installations for microgrids are high in Southern Africa compared to other regions [6,9], thus the economic factors in the region favours the cheaper alternative which is diesel power generation. However, the variability in diesel prices due to internal and external markets makes power generation with diesel power plants expensive in Zambia. For example, due to the conflict in Ukraine,

there are instabilities in the energy sector which has brought up high energy/fuel prices across the globe, consequently increasing the on-site cost of diesel for rural electrification in Zambia. Sensitivity analysis of fuel prices shows that a change in the cost of diesel fuel by \$ 1 changes the Levelized cost of electricity by \$ 0.283. This shows that generation with diesel for microgrids in rural areas is not the best option especially that most of these areas do not have much economic activities to absorb the variability in energy prices and justify the high cost of energy. Therefore, as shown increasing penetration of Solar in the operation of the power plant lowers and stabilizes the LCoE and will ease power planning despite the high investment costs.

Additionally, the implementation of scenarios with DG would face strict measures from the Zambia Environmental Management Agency (ZEMA) in the approval process of their Environmental Social Impact Assessment (ESIA) that is needed before the construction of such projects in Zambia. The simulation results presented at the 13th international conference on sustainable energy and environmental protection [43] show that the carbon dioxide and other emissions were lowest in the 100% PV and storage system compared to the other three schemes. Therefore, despite the high investment cost, a PV with a battery offers environmental above the other three schemes.

Another aspect of the options presented is in the LCoE. Each of the scenarios resulted in different values of LCoE. One important aspect is that adding the loading and change in capacities did not alter the LCoE for C and D scenarios. Only the initial cost increased for each of the scenarios. Scenarios C and D had the low LCoE among all the scenarios. Scenario C, which has DG, has the second lowest LCoE (minimal difference). Its carbon emissions are lower than the other two options with DG added. It is the second attractive option in line with COP26 guidelines. The other two options, scenarios A and B, have high LCoE, consequently affecting the number of customers connected to the power grid. Additionally, their emissions also entail that this is a non-attractive option.

The use of DG entails purchasing diesel at a particular cost. Scenario A has the highest Diesel usage, and the cost is high. Operations with a DG will lead to more demand for diesel, and the sensitivity analysis of the variation in the cost of diesel shows an increase in LCoE in Zambia [39]. Besides the emissions into the atmosphere, Zambia is not a producer of petroleum products. The addition of Solar PV causes a reduction in the number of operational hours of DG. There is a resulting decrease in the cost and consumption of diesel.

The inclusion of the battery bank made the systems more reliable. Additionally, the PV array provided most of the power to feed the load. Therefore, this scheme proves to be financially sustainable in the long run. In addition, once installed, the batteries, converter, and PV array would have minimal expenses associated with operation and maintenance.

Moreover, the generator's life would be higher for the PV-Diesel hybrid system than any other three schemes. In line with [52], a PV-Diesel hybrid system with storage can reduce diesel consumption emissions. When the PV power output in the hybrid system cannot meet the required load demand during cloud cover or on rainy days, the battery will sustain the critical load (battery autonomy) without charging, similar findings to Ref. [43]. Furthermore, the solar irradiation and the rising fuel prices continuously are the factors that would offer an economic advantage to the PV-Diesel Hybrid system compared to a stand-alone diesel generator system, as reported by Ref. [53]. However, accounting for the storage with the load than the PV capacity results in high installation costs. Additionally, there is a very high excess of electricity produced.

The application of pumped hydro with renewable energy (wind or solar) is far from reaching the implementation stage and has been proposed as a future research direction (just recently) that will be interesting to explore further [27]. The assessment of potential is done for rivers and shores in Ref. [28]. The latter is one important aspect. When a river or dam is not near the implementation place, extensive hydraulic

structures are needed for the overall arrangement of the system [26,27]. In this study, the landscape does not favour hydro storage and the practical and viable solution that is capable of immediate implementation is PV with battery storage. Even then emphasis is place to explore the potential of hybrid microgrids with pumped hydro storage in regions with potential for such since overtime the system is more cost effective than PV with batteries.

Another alternative option for Chilubi Island is to add wind power resources to evaluate the possible production as a hybrid option. Measurements at 8 locations in Zambia have been done, but the closest to Chilubi was 300 km, making the system competitive with extending the national grid. Thus, discussions within the utility company should focus on integrating wind power with the national grid and supplying rural areas like the Chilubi district. Thus, the estimated cost of connecting Chilubi and having wind power is higher than the options obtained in the results.

6. Conclusion

The study focused on the viability of the PV-Diesel hybrid system for rural electrification with the case study of the Chilubi district in Zambia. The cost of extending the grid to rural areas is very high in Zambia and in Southern Africa and this can be avoided by implementing distributed microgrids particularly for rural electrification. The results herein show that the stand-alone diesel generating sets system is not economically sustainable as they have the highest LCoE due to high diesel consumption despite having the lowest investment costs. Additionally, the scheme has the highest carbon emissions, which contradicts the goals of COP26 on greenhouse gas emissions from energy sources. From the results, the PV-Diesel hybrid system with battery gives the lowest LCoE out of the four considered systems. However, the study also shows that the capital cost of PV power installations for microgrids are expensive in Zambia compared to other developing regions. There is a need for deliberate political drive and policies to increase penetration or installation of PV hybrid systems with a larger share from renewables. This is needed because diesel power generation are a quick fix that have higher operational costs and environmental impact even though they are cheaper to setup than the counter part of PV-hybrid systems. This drive should also extend to include integration of PV resources into existing diesel power plants because the electricity generation costs get lower by integrating renewable sources into the microgrid, making electricity cheap and affordable in the long run. This result is competitive in terms of energy cost, especially in areas with low economic activities and investment incentives in energy projects. Additionally, the periodic fluctuation of diesel pump prices increases the uncertainty of system operation and energy costs. Therefore, implementing PV with storage ensures energy security and is an attractive option as the cost of energy is very low and thus is unique for rural areas with less economic activities. To satisfy government targets and the universal right of having access to green energy, microgrids can offer cheaper power generation for rural electrification. In certain regions like Zambia and Southern Africa, a strong political will that push for deliberate policies or subsidies that favour implementation and installation of power generation plants with predominantly renewable energy resources may be needed to dis-incentivise the alternative of cheaper of diesel power plants. Overall, solar is the best solution for Sub-Saharan Africa's energy deficiency, especially since there is a continued downward trend in the cost of PV per kWp.

Author contributions

Mulenga Enock (ME), Mupeta Henry (MH), Kabanshi Alan (KA): Conceptualisation, Methodology, Software. **KA, MH and Nyirenda Elvis (NE):** Simulations. **ME, KA, HM, NE:** Analysis, Visualization, Investigation. **MH, ME, KA:** Writing of manuscript. **ME, KA, Ndiaye Musa (NM) and Mulenga Kabwe (MK):** rewrite and editing. **ME, KA,**

NE: Supervision KA.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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